

ASAD ABDULLAH

Senior Mobile App Developer — Native Android & Cross-Platform Flutter — Kotlin · Dart · Jetpack Compose

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Professional Summary

Senior Mobile App Developer with **4+ years** of experience delivering production-grade **native Android and cross-platform Flutter applications** across **fintech, VoIP, e-commerce, navigation, and eSIM** domains. Expert in **Kotlin, Jetpack Compose, Flutter/Dart for hybrid mobile development**, and **Edge AI / on-device LLM inference** using Google MediaPipe, Gemma/Llama, and Qwen 2.5. Architected privacy-first AI companions with **RLHF (Reinforcement Learning from Human Feedback)** and vector-based long-term memory, custom RAG engines, hybrid OCR pipelines, and SMS parsing systems. Proven integrator of **WebRTC, Web3/MetaMask, and eSIM lifecycle management**. Built scalable **cross-platform mobile solutions** reducing development cycles while maintaining native performance. Multiple award recipient at ArhamSoft Pvt. Ltd. for technical excellence and dedication.

Technical Skills

Languages: Kotlin, Dart, Java, SQL, Python

Cross-Platform / Hybrid Development: Flutter/Dart, Riverpod v2, BLoC Pattern, Provider, GetX, flutter_bloc, Material Design M2/M3, Platform Channels (MethodChannel, EventChannel), Native Module Integration, Responsive UI Design, Supabase, Firebase for Flutter

Native Android Development: Android SDK, Jetpack Compose, Jetpack Components (ViewModel, Navigation, WorkManager, Room), Hilt/Dagger, Coroutines/Flow, Retrofit, Coil/Glide/Picasso

AI / ML: Google MediaPipe GenAI (LlmInference), Gemma/Llama Models, Qwen 2.5 LLM, On-Device Inference, RAG (Retrieval-Augmented Generation), RLHF (Reinforcement Learning from Human Feedback), Vector Embeddings, Semantic Search, Google ML Kit, TensorFlow Lite, Model Quantization, Prompt Engineering, SMS Parsing

Specialized Integrations: WebRTC, JANUS Server, Web3/MetaMask/WalletConnect, eSIM (RSP/GSMA), Firebase (Auth, Firestore, FCM, Crashlytics), Google Maps API, Payment Gateways, Push Notifications

Architecture & Design Patterns: MVVM, MVP, Clean Architecture, Repository Pattern, SOLID Principles, DRY, KISS, MVI, Dependency Injection

Databases & Storage: Room (SQLite), Drift (Flutter), Firestore, Supabase (PostgreSQL), Hive, SharedPreferences, Datastore

DevOps / Tools: Git, GitHub, GitLab, CI/CD (GitHub Actions, Codemagic, Bitrise), Android Keystore, Biometrics, RESTful APIs, GraphQL, OkHttp, Dio, Jsoup, WhatsApp Chat Parsing, Speech-to-Text (STT), App Store Deployment (Google Play, App Store), Firebase Crashlytics, Performance Monitoring

Experience

ArhamSoft Pvt. Ltd.

Apr 2023 – Present

Senior Android Application Developer

Lahore, Pakistan

- Architect and ship production-grade **native Android and cross-platform Flutter applications** across **navigation, fintech, VoIP, and e-commerce** domains serving thousands of active users.
- Lead technical decision-making for **hybrid vs. native architecture**, optimizing development velocity while maintaining performance and UX quality across iOS and Android platforms.
- Designed a **Privacy-First On-Device AI Assistant** using **Google MediaPipe + Qwen 2.5 LLM** — 100% local inference, zero cloud dependency, eliminating latency and API costs.
- Built a **Custom RAG Engine** injecting real-time SQLite financial snapshots (spending trends, subscriptions, top merchants) into LLM prompts for hyper-personalized insights.
- Developed a **Hybrid Receipt OCR Pipeline** combining **ML Kit + local LLM** to extract structured JSON (merchants, items, totals, tax) from physical receipts with automatic DB merging.
- Engineered a **SMS Parsing Engine** supporting **46+ global bank formats** via background **WorkManager**, generating auto-timestamped calendar notes from banking notifications with zero battery drain.
- Integrated **WebRTC + JANUS Server** for a real-time VoIP call center app, achieving **sub-100ms call setup times** with WebSocket-based signaling.
- Implemented **eSIM lifecycle management** — browse, purchase, and activate international data plans without a physical SIM, integrated with **Web3/MetaMask/WalletConnect** crypto payments.
- Reduced redundant API calls via a **local event-driven architecture**, improving network efficiency and app performance measurably.
- Led **Kotlin, Jetpack Compose, Flutter/Dart, MVVM, Clean Architecture** migrations, improving code maintainability by an estimated **30%**; mentored junior developers and enforced SOLID/DRY/KISS in code reviews.

ArhamSoft Pvt. Ltd.

Sep 2022 – Mar 2023

Flutter Developer — Cross-Platform Mobile Development

Lahore, Pakistan

- Developed and maintained **cross-platform hybrid mobile applications** using Flutter/Dart, accelerating delivery timelines by enabling simultaneous iOS and Android deployment from a single codebase.
- Implemented **native platform integrations** via Platform Channels (MethodChannel, EventChannel) for features requiring device-specific APIs, maintaining near-native performance.
- Collaborated with UI/UX designers to build pixel-perfect, responsive interfaces using **Material Design M2/M3** components, ensuring visual consistency and accessibility across Android and iOS platforms.
- Architected state management solutions using **BLoC, Riverpod, and Provider** patterns, ensuring scalable and maintainable code architecture for complex hybrid applications.
- Integrated Firebase services (Auth, Firestore, FCM) and Supabase backend for real-time data synchronization across mobile and web platforms.

ArhamSoft Pvt. Ltd.

Oct 2021 – Aug 2022

Junior Android Developer

Lahore, Pakistan

- Contributed to Android module development and bug-fixing within production apps, improving stability and UX.
- Gained production proficiency in **Android SDK, Jetpack Components, REST API integration, and Firebase.**

Projects

AuraChat — AI Companion with RLHF | *Kotlin, MediaPipe GenAI, Vector Embeddings, RAG, RLHF, Jetpack Compose*

- **100% offline AI companion** with on-device LLM inference (Gemma/Llama via MediaPipe) — zero data leaves device, guaranteeing privacy.
- Implemented **Real-Time RLHF (Reinforcement Learning from Human Feedback)** — user corrections update episodic memory vector embeddings and refine semantic knowledge base to prevent recurring errors.
- Built **Vector-Based Long-Term Memory** using local vector storage and semantic search for persistent context recall across sessions.
- Engineered **WhatsApp Persona Mirroring** — custom parser analyzes exported chat ‘.txt’ files to clone real-world communication styles into AI personas.
- Designed premium **glassmorphic UI** with bento-box dashboards, tactile voice UX (STT + waveforms), and dynamic LLM parameter tuning studio.

Budget — AI-Powered Fintech App | *Kotlin, MediaPipe GenAI, Qwen 2.5, ML Kit, RAG, Room, WorkManager*

- Privacy-first on-device AI assistant with **100% local LLM inference** using MediaPipe + Qwen 2.5, zero cloud dependency.
- Custom **RAG layer** synthesizing real-time SQLite financial data (income/expense, categories, subscriptions) into LLM context.
- Hybrid OCR pipeline (ML Kit + LLM) extracting structured JSON from physical receipts with automatic DB merging.
- SMS-to-Note automation engine parsing **46+ bank formats** via WorkManager; persistent multi-turn chat with sliding window context management.

SpeakerBus — VoIP Call Center App | *Kotlin, WebRTC, JANUS Server, WebSockets, Android SDK*

- Scalable VoIP call center solution with **WebRTC + JANUS**, achieving **sub-100ms call setup times** and high-concurrency call operations.
- WebSocket-based signaling architecture for seamless bidirectional data transmission.

Travel eSIM App | *Kotlin, Jetpack Compose, MVVM, Clean Architecture, MetaMask, WalletConnect*

- Full-featured **eSIM management app** — browse, purchase, and activate international data plans without a physical SIM.
- **Web3 crypto payment flow** via MetaMask/WalletConnect; local event-driven architecture eliminating redundant API calls.

PixelDrape — Flutter Wallpaper App (Hybrid) | *Flutter, Riverpod v2, Supabase, MVVM, Clean Architecture*

- **Cross-platform hybrid wallpaper application** with admin-curated content delivery via Supabase backend, deployed simultaneously to iOS and Android.
- Dark glassmorphism UI with responsive design patterns ensuring pixel-perfect rendering across device form factors and orientations.
- Implemented MVVM/Clean Architecture with Riverpod v2 state management, maintaining 95%+ code reusability between platforms.

Blight — Food Freshness Classifier (Hybrid) | *Flutter, TFLite, flutter_bloc, Drift ORM, Clean Architecture*

- **Cross-platform on-device ML application** using TensorFlow Lite for real-time food freshness classification, running identically on iOS and Android.
- Architected with BLoC pattern and Drift ORM for local-first data persistence; Clean Architecture ensures testability and scalability.

KIOSKS — In-Store Shopping Companion | *Kotlin, Google Maps API, Android SDK, RESTful APIs*

- In-store navigation app with product search and step-by-step routing to shelf locations using Google Maps API.

University Bus Tracking App | *Kotlin, Google Maps API, Firebase Firestore, FCM, Real-time GPS*

- Real-time student transportation platform with live GPS tracking, driver-drawn route updates, automated attendance, and fee receipt generation.

Education

Government College University, Faisalabad
Bachelor of Science in Computer Science (BS-CS)

2016 – 2020
Faisalabad, Pakistan

Awards & Recognition

The Shining Star & The Rising Star — ArhamSoft Pvt. Ltd. (Mar 2024)
The Shining Star & Unmatched Dedication — ArhamSoft Pvt. Ltd. (Mar 2022)
The Rising Star — ArhamSoft Pvt. Ltd. (Aug 2022)

Profiles

GitHub: github.com/AsadAhmad01
Portfolio: linktr.ee/asad..abdullah